



10104 - The Fastest Protostellar Jet

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCam imaging				
	1	FeII, H2, Paschen-alpha, Bracket-alpha	NIRCam Imaging	(2) GGD27-JET-North
	2	FeII, H2, Paschen-alpha, Bracket-alpha	NIRCam Imaging	(1) GGD27-JET-South

ABSTRACT

The birth of massive stars remains one of the least understood, but one of the most important aspects of star formation. NIRCam images will trace the interaction of the fastest known jet from a forming star with the surrounding molecular cloud. This 1,200 km/s jet emerges from IRAS 18162-2048, an embedded ~20 Solar mass young stellar object surrounded by a 5 Solar mass disk at a distance of 1.4 kpc. This jet powers a group of red-shifted, high-excitation shocks, HH 80 and 81, located 1.5 pc to the south-southwest. We propose narrow-band JWST/NIRCam imaging to study the environment of IRAS 18162-2048 where the jet breaks out of the parent cloud core, the HH 80/81 shocks, and the recently discovered, blue-shifted counterflow where the jet burrows into the parent cloud. H-recombination lines and [FeII] trace the jet and hard shocks where faster ejecta overtakes

slower material. Molecular hydrogen and vibrationally-excited CO traces the interface where momentum is transferred from the jet to the host cloud. Extinction corrected fluxes and emission-line proper motions will be used to estimate densities and mass-loss-rates. JWST resolution will probe the collimation region where a wind from the source is converted into a jet. The narrow-band NIRCam filters will characterize photon-dominated-regions, the near-IR luminosity function of the host-cluster and, using evolutionary models, the masses and ages of the stars. These observations will shed light on the conditions which lead to the production of one of the most extreme protostellar outflows known in the Galaxy, probe the physics of outflow-driven feedback, and serve as a template for massive star formation.

OBSERVING DESCRIPTION

NIRCam will be used to image a 2.6×12 arcminute field with two pointings, GGD27-JET-South and GGD27-JET-North. We will use the FULLBOX primary dither mode. The mosaic has an Orientation requirement so that the full spatial extent of the near-IR emission from the GGD 27 reflection nebulae around IRAS 18162-2048 and most of the spatial extent of the outflow is well covered.

To make the observations possible to schedule (e.g. observing the Micrometeoroid Impact constraint), we use an orientation requirements in which the major dimension of the NIRCam grids are misaligned with the jet axis. Thus there is a 'zig-zag' between the two pointings.

In the GGD27-JET-South pointing, the mosaic is oriented so that the jet source, IRAS 18162-2048, is located near the northeast corner of the southern imaged field while the high-excitation HH 80/81 shocks are in the southeast corner.

For each pointing, in each narrow-band filter, two integrations with 5 groups per integration are combined to a total exposure time of 1932 seconds. We use the SHALLOW2 readout mode with 4TIGHT primary dithers.

For the medium band filters, used to remove continuum in the narrow-bands, we use 2 groups per integration for a total exposure time of 644 second.

The filters used are: F187N and F405N, F212N and F470N, F164N and F323N, and F210M and F460M in both the GGD27-JET-South and GGD27-JET-North pointings (visits 1 and 2).

Proposal 10104 - Targets - The Fastest Protostellar Jet

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	GGD27-JET-South	RA: 18 19 9.7846 (274.7907692d) Dec: -20 49 41.00 (-20.82806d) Equinox: J2000	Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Parallax: 0.00139" Epoch of Position: 2000	
	Comments: Category=Star Description=[Circumstellar matter, Emission line stars, O dwarfs, Pre-main sequence stars, Protostars] Extended=YES				
Fixed Targets	(2)	GGD27-JET-North	RA: 18 19 15.4277 (274.8142821d) Dec: -20 44 1.00 (-20.73361d) Equinox: J2000	Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Parallax: 0.00139" Epoch of Position: 2000	
	Comments: Category=Star Description=[Circumstellar matter, Emission line stars, O dwarfs, Pre-main sequence stars, Protostars] Extended=YES				

Proposal 10104 - Observation 1 - The Fastest Protostellar Jet

Observation	Proposal 10104, Observation 1: FeII, H2, Paschen-alpha, Bracket-alpha Wed May 13 21:00:20 GMT 2026								
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging								
Diagnosics	(FeII, H2, Paschen-alpha, Bracket-alpha (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(2)	GGD27-JET-North	RA: 18 19 15.4277 (274.8142821d) Dec: -20 44 1.00 (-20.73361d) Equinox: J2000		Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Parallax: 0.00139" Epoch of Position: 2000				
	Comments: Category=Star Description=[Circumstellar matter, Emission line stars, O dwarfs, Pre-main sequence stars, Protostars] Extended=YES								
Template	Module		Subarray		Target Placement				
	ALL		FULL		Module gap (large extended source)				
Dithers	#	Primary Dither Type	Primary Dithers		Subpixel Dither Type	Dither Size		Subpixel Positions	
	1	FULLBOX	4TIGHT		STANDARD			1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time
	1	F212N	F470N+F444W	SHALLOW2	5	2	8	4	1932.619
	2	F187N	F405N+F444W	SHALLOW2	5	2	8	4	1932.619
	3	F164N+F150W2	F323N+F322W2	SHALLOW2	5	2	8	4	1932.619
	4	F210M	F460M	SHALLOW2	2	2	8	4	644.206
	5	F182M	F360M	SHALLOW2	2	2	8	4	644.206
Special Requirements	Aperture PA Range 96 to 96 Degrees (V3 96.07457694 to 96.07457694)								

Proposal 10104 - Observation 2 - The Fastest Protostellar Jet

Observation	Proposal 10104, Observation 2: FeII, H2, Paschen-alpha, Bracket-alpha Wed May 13 21:00:20 GMT 2026									
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging									
Diagnostics	(FeII, H2, Paschen-alpha, Bracket-alpha (Obs 2)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous			
	(1)	GGD27-JET-South	RA: 18 19 9.7846 (274.7907692d) Dec: -20 49 41.00 (-20.82806d) Equinox: J2000		Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Parallax: 0.00139" Epoch of Position: 2000					
Template	<i>Comments:</i> <i>Category=Star</i> <i>Description=[Circumstellar matter, Emission line stars, O dwarfs, Pre-main sequence stars, Protostars]</i> <i>Extended=YES</i>									
	Module	Subarray		Target Placement						
Dithers	ALL		FULL		Module gap (large extended source)					
	#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions				
Spectral Elements	1	FULLBOX	4TIGHT	STANDARD	1					
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
Special Requirements	1	F212N	F470N+F444W	SHALLOW2	5	2	8	4	1932.619	
	2	F187N	F405N+F444W	SHALLOW2	5	2	8	4	1932.619	
	3	F164N+F150W2	F323N+F322W2	SHALLOW2	5	2	8	4	1932.619	
	4	F210M	F460M	SHALLOW2	2	2	8	4	644.206	
	5	F182M	F360M	SHALLOW2	2	2	8	4	644.206	
Special Requirements	Aperture PA Range 96 to 96 Degrees (V3 96.07457694 to 96.07457694)									